

THE HONG KONG POLYTECHNIC UNIVERSITY
DEPARTMENT OF COMPUTING
EXAMINATION

Course : Broad Discipline of COMP-61431, DD in COMP & Mgt-61432

Subject : COMP4431 Artificial Intelligence

Group : 1011

Session : 2015 / 2016 Semester I

Date : 12 December 2015

Time : 12:30-14:30

Time Allowed: 2 Hours

Subject Lecturer: Dr LIU Yan

This question paper has ____4____ pages.
(Some pages may be intentionally omitted.)

Instructions to Candidates:

1. Close book.
2. Calculator is permitted.
3. Answer ALL questions.

Do not turn this page until you are told to do so!

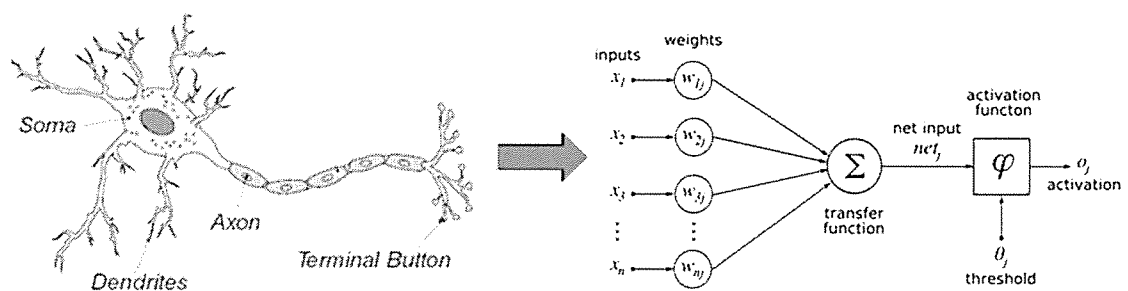
Section B**[80 Points]**

Question 2. Answer the question in a short paragraph.

2.1. Explain the differences between hill-climbing search and simulated annealing search using the 8-queen problem shown below. The number in each block indicates the pair of attacks when queen in the same column moves to that block. [4]

18	12	14	13	13	12	14	14
14	16	13	15	12	14	12	16
14	12	18	13	15	12	14	14
15	14	14	♂	13	16	13	16
♂	14	17	15	♂	14	16	16
17	♂	16	18	15	♂	15	♂
18	14	♂	15	15	14	♂	16
14	14	13	17	12	14	12	18

2.2. Could you explain how artificial neuron simulates biological neuron according to the figures shown below? [4]

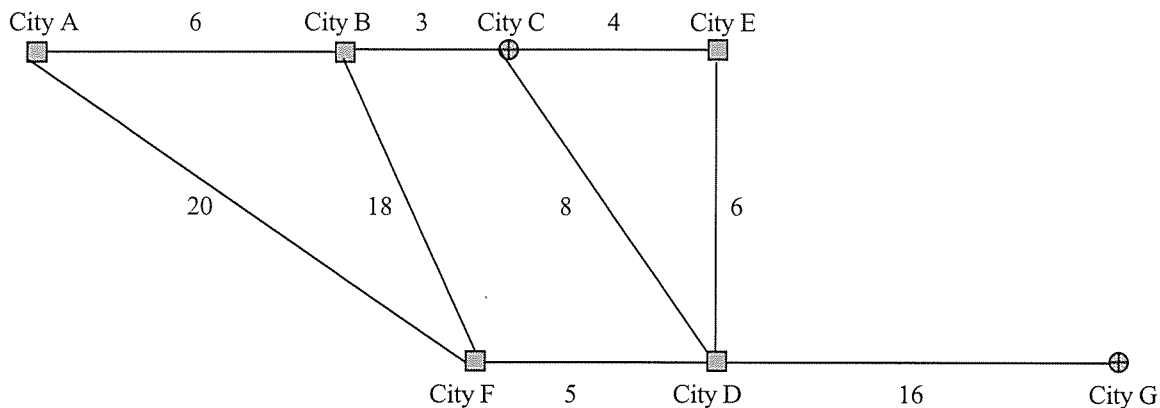


2.3. Could you please explain what is probably approximately correct learning? What is the sample complexity of the hypothesis space? [6]

2.4. We want to classify the 50,000 data to two categories of class A and class B. 20,000 data are used for training and other 30,000 data are used for test. Linear support vector machine is used as the classifier. Each data includes 1000 features. We want to select the most important 100 features for classification. Please explain how to utilize genetic algorithm for feature selection of this application. [8]

Question 3. We want to find a way to drive from City C to City G. We show the distance between two cities in the figure. We formulate the problem to the search tree. The heuristic information is provided as the straight line distance between two cities as shown below:

A-G: 30; B-G: 25; C-G: 20; D-G: 15; E-G: 10; F-G: 5; G-G:0



3.1. Solve the problem using breadth-first search. The target is to find a path from City C to City G. Draw the search tree and explain the order of the node checked or/and expanded. [4]

3.2. Solve the problem using depth-first search. The target is to find a path from City C to City G. Draw the search tree and explain the order of the node checked or/and expanded. [4]

3.3. Solve the problem using uniform-Cost search. Draw the search tree and explain the order of the node checked or/and expanded. [4]

3.4. Solve the problem using greedy best-first search. Draw the search tree and explain the order of the node checked or/and expanded. [4]

3.5. Solve the problem using A* search. Draw the search tree and explain the order of the node checked or/and expanded. [4]

Question 4. The knowledge base is given as the set of sentences. Please use the knowledge based shown below to illustrate the difference between forward chaining and backward chaining when Q2 is the query. [8]

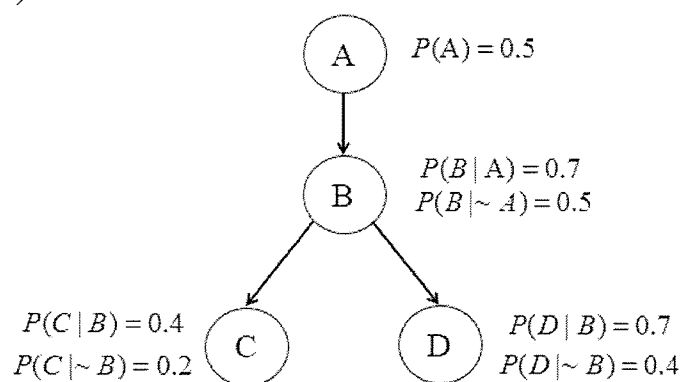
- A
- B
- C

- $A \wedge B \Rightarrow D$
- $B \wedge C \Rightarrow E$
- $A \wedge C \Rightarrow F$
- $A \wedge F \Rightarrow G$
- $D \wedge F \Rightarrow K$
- $G \wedge K \Rightarrow Q1$
- $E \Rightarrow H$
- $H \wedge C \Rightarrow Q2$

Question 5. We make the decision of "Yes" or "No" using a decision tree. Could you please construct a decision tree according to the given training dataset? Please write down every step of attribute selection. [16]

DATA	F1	F2	F3	F4	F5	LABEL
1	F1V2	F2V1	F3V1	F4V1	F5V2	YES
2	F1V2	F2V2	F3V1	F4V1	F5V2	NO
3	F1V2	F2V2	F3V1	F4V1	F5V2	NO
4	F1V2	F2V1	F3V1	F4V3	F5V2	NO
5	F1V1	F2V1	F3V1	F4V3	F5V2	YES
6	F1V1	F2V2	F3V1	F4V2	F5V2	YES
7	F1V1	F2V2	F3V1	F4V2	F5V2	YES
8	F1V1	F2V1	F3V1	F4V1	F5V2	YES
9	F1V1	F2V2	F3V1	F4V1	F5V2	NO
10	F1V1	F2V2	F3V1	F4V1	F5V2	NO
11	F1V2	F2V2	F3V2	F4V3	F5V1	NO
12	F1V2	F2V2	F3V2	F4V3	F5V1	NO
13	F1V1	F2V2	F3V2	F4V2	F5V1	YES
14	F1V1	F2V2	F3V2	F4V3	F5V1	YES
15	F1V2	F2V1	F3V2	F4V3	F5V1	NO
16	F1V2	F2V1	F3V2	F4V3	F5V1	NO
17	F1V1	F2V1	F3V2	F4V2	F5V1	YES
18	F1V1	F2V1	F3V2	F4V2	F5V1	YES
19	F1V1	F2V1	F3V2	F4V1	F5V1	YES
20	F1V1	F2V2	F3V2	F4V1	F5V1	NO

Question 6. According to the Bayesian Networks shown below, please calculate $P(\sim C)$ and $P(B|D)$. [14]



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